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Kampmann Lab iNeuron pre-differentiation & differentiation protocol

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*Equal contribution, listed alphabetically

This protocol describes differentiation of iPSCs with stably integrated doxycycline-inducible Ngn2 (such as i3Ns).

Reference:

Tian et al (2019). CRISPR Interference-Based Platform for Multimodal Genetic Screens in Human iPSC-Derived Neurons. Neuron pii: S0896-6273(19)30640-3. [Epub ahead of print]
PubMed PMID: 31422865.

Culture iPSCs

1. Culture iPS cells with ROCK inhibitor only when plating or when colonies constitute fewer than about 20 cells
2. Generally, change the media every other day, unless cells are more than 50% confluent, change StemFlex every day
3. Passage cells when 60-75% confluent. Should never be over 90% confluent

Cell Culture Vessel	Surface area (cm ²)	Scaling factor from 12-well plate	Matrigel Volume	DPBS for Wash	Accutase Volume	DPBS Volume for Accutase Dilution	Media Volume	Recommended Day -3 Seeding Density for pre-differentiation*	Recommended Day 0 Seeding Density for PDL-Coated Vessels (Connor)	Recommended Day 0 Seeding Density for PDL-Coated Vessels (Ruilin)
96-well Plate	0.32	0.09	50uL	100uL	30uL	150uL	100uL	-	1 x 10 ⁴	0.1-0.25 x 10 ⁴
24-well Plate	1.9	0.54	250uL	500uL	200uL	1mL	500uL	-	1 x 10 ⁵ (4 x 10 ⁴ for imaging)	0.1 x 10 ⁶
12-well Plate	3.5	1.00	500uL	1mL	300uL	1.5mL	1mL	-	2 x 10 ⁵	0.2 x 10 ⁶
6-well Plate	9.6	2.74	1mL	2mL	500uL	2.5mL	2mL	7.5 x 10 ⁵	5 x 10 ⁵	1-2 x 10 ⁶
10-cm Dish	56.7	16.20	4mL	10mL	2mL	10mL	10mL	4 x 10 ⁶	7 x 10 ⁶	7-10 x 10 ⁶
T175 Flask	175	50.00	8mL	20mL	5mL	25mL	25mL	-	2 x 10 ⁷	-
15cm Dish	145	41.43	12	35	6	30	35	-	21 x 10 ⁶	20-28 x 10 ⁶

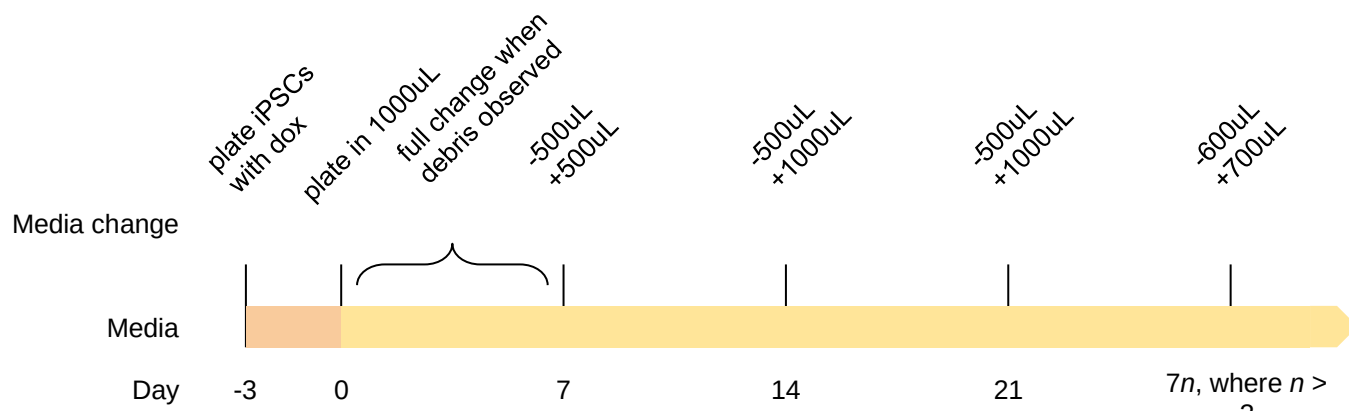
*Remember- you will replate on Day 0, so make sure you have enough pre-differentiated cells. Day -3 seeding has nothing to do with the format of seeding on Day 0, other than that you need enough cells.

Pre-Differentiation and Differentiation Notes:

1. Keep reagents in 4°C for a maximum of 4 weeks, If needing long term storage, don't freeze thaw more than 3 times.

2. Sometimes pre-differentiated cells take a long time to come off of the plate on Day 0. This is usually okay. It is important that your cells are single cells.

3. Feeding schedule and volumes:



Days -3-0: Pre-differentiation

Make N2 Pre-differentiation Media:

Component	Stock Concentration	Final Concentration	Dilution Factor	For 100 mL (mL)	For 50 mL (units in cell)
Knockout DMEM/F12	1X	1X	1	100	50mL
NEAA	100X	1X	100	1	500uL
N2 Supplement	100X	1X	100	1	500uL
NT-3	10ug/mL	10ng/mL	1000	0.1	50uL
BDNF	10ug/mL	10ng/mL	1000	0.1	50uL
Mouse Laminin*	1.0mg/mL	1ug/mL	1000	0.1	50uL
ROCK Inhibitor**	10uM	10nM	1000	0.1	50uL
Doxycycline***	2mg/mL	2ug/mL	1000	0.1	50uL

*Mouse Laminin comes at variable concentrations, so make sure the final concentration is correct in the media you are making!

**Only add during plating of iPSCs on Day -3 and omit for any further media changes

***Add immediately before use

Protocol:

1. Coat plate with matrigel diluted in Knockout DMEM. Coat for at least 30 minutes (or O/N). Coated plates can last in the 37°C incubator for 14 days
2. Aspirate media from iPSCs and wash with DPBS
3. Add accutase and incubate at 37°C for 3 minutes; if necessary, incubate for additional time up to 7 minutes
4. Use gentle agitation to release cells, and collect in an Eppendorf tube or conical with DPBS
5. Spin cells at 200 x g for 5 minutes, resuspend in N2 pre-diff media
6. Count cells and add desired amount to an Eppendorf tube or conical, spin, resuspend in N2 pre-differentiation media, and plate onto Matrigel-coated cell culture vessels for the pre-differentiation. Reference seeding density chart for plating.
7. Perform ½ pre-differentiation media change every day (w/dox, no RI) throughout days 0-3 pre-differentiation period or full media change on day -1 or day -2. Be consistent.
8. Going into differentiation, it is best if cells are in single cell suspension. Optional: use cell strainer
9. Pre-differentiated cells can be frozen in 10%DMSO + N2/B27 Differentiation media (**can also freeze in pre-diff media)

Day 0: Releasing and Plating Pre-Differentiated iNeurons

Make N2/B27 Differentiation Media:

Component	Stock Concentration	Final Concentration	Dilution Factor	For 100mL (mL)	For 50 mL (units in cell)
DMEM/F12*	1X	0.5X	2	50	25mL
Neurobasal-A	1X	0.5X	2	50	25mL
NEAA	100X	1X	100	1	500uL
GlutaMAX	100X	0.5X	200	0.5	250uL
N2 Supplement	100X	0.5X	200	0.5	250uL
B27-VA Supplement	50X	0.5X	100	1	500uL
NT-3	10ug/mL	10ng/mL	1000	0.1	50uL
BDNF	10ug/mL	10ng/mL	1000	0.1	50uL
Mouse Laminin**	1.0mg/mL	1ug/mL	1000	0.1	50uL
Doxycycline***	2mg/mL	2ug/mL	1000	0.1	50uL

*DMEM/F12 with either bicarbonate or HEPES buffer is okay, but we have had qualitatively better neurons with HEPES buffer when grown and differentiated side-by-side. HEPES is essential for imaging, as media will change color after ~30 minutes if using bicarbonate buffered media.

**Mouse Laminin comes at variable concentrations, so make sure the final concentration is correct in the media you are making!

***Add immediately before use and only on Day 0

1. Aspirate media and wash with DPBS
2. Add accutase and incubate at 37°C for 3 minutes; if necessary, incubate for additional time up to 7 minutes
3. Use gentle agitation to release cells, and collect in an Eppendorf tube or conical with DPBS
4. Spin cells at 200 x *g* for 5 minutes
5. Carefully aspirate DPBS/accutase solution and resuspend in Classic N2/B27 Differentiation Media
6. Count cells, dilute to appropriate density, and plate onto PDL-coated cell culture vessels
7. Leftover cells can be frozen down in pre-differentiation media + 10% DMSO

Day 0+: Neuronal differentiation

1. Full media change on day 3 post plating, once debris is observed.
2. Pipet at the wall of the plate. DO NOT TOUCH BOTTOM OF PLATE! Be extra careful with 96 well plates
3. Change Classic N2/B27 Differentiation media at minimum once every week without dox or rock inhibitor.
4. Closely monitor. Wait for a week at minimum before using for experiment. Preferably, use after 2 weeks.

Component	Vendor/Manufacturer	Catalog Number	Volume/Amount
Matrigel Matrix	Corning	356231	10mL
Knockout DMEM	Thermo Fisher	10829-018	500mL
Essential 8 Medium	Thermo Fisher	A1517001	500mL base. 10mL 50X supp
ROCK inhibitor (Y-27632 dihydrochloride)	Tocris	125410	10mg
StemPro Accutase Cell Dissociation Reagent	Thermo Fisher	A11105-01	100mL
DBPS (without Ca ²⁺ or Mg ²⁺)	multiple	-	-
Knockout DMEM/F12	Thermo Fisher	12660-012	500mL
DMEM/F12	Thermo Fisher	11320-033	500mL
Neurobasal-A	Thermo Fisher	10888-022	500mL
BrainPhys	Stem Cell	05790	500mL
NEAA (MEM Non-Essential Amino Acids)	Thermo Fisher	11140-050	100mL
GlutaMAX	Thermo Fisher	35050-061	100mL
N2 Supplement	Thermo Fisher	17502-048	5mL
B27 -VA Supplement	Thermo Fisher	12587010	10mL
NT-3	PeproTech	450-03	10ug/50ug
BDNF	PeproTech	450-02	10ug/50ug
Mouse Laminin	Thermo Fisher	23017-015	1mg
Doxycycline hydrochloride	Sigma-Aldrich	D3447-500MG	500mg